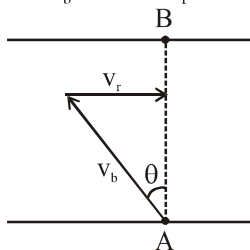


Solution :-

- As the train is moving, so the person in the train observe the path of the stone as parabolic path, because according to him the stone has both horizontal and vertical velocities. Horizontal component is due to relative motion w.r.t. train and vertical component is provided by the person on ground.
- For a particle moving in a circle with constant angular speed velocity vector is always tangential to the circle and the acceleration vector always points towards the centre of circular path and along the radius of the circle. Since, tangential vector is perpendicular to radial vector, therefore velocity vector will be perpendicular to the acceleration vector, but in this case acceleration vector is not tangential to the circle.
- For the boy to reach at point B from point A : $\rightarrow v_b \sin \theta = v_r$



$v_b \cos \theta$ has to be non-zero for the displacement along the y-direction.

$$\text{So, } v_b = \frac{v_r}{\sin \theta}$$

here, θ has to be less than 90° , so $\sin \theta < 1$ for $\theta < 90^\circ$

hence, $v_b > v_r$

- (1) False : energy is not conserved during inelastic collision.
 - (2) False : Temperature can be negative
 - (3) False : Gravitational potential energy vary from point to point in space
 - (4) True : Scalar quantity is independent from direction hence has the same value for observers with different orientations of the axes.

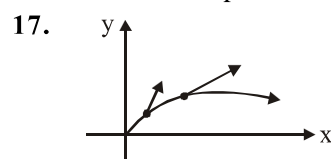
- As given motion is two dimensional motion and given that instantaneous speed v_0 is positive constant. Acceleration is rate of change of velocity, hence it will also be in the plane of motion.
- Let $\vec{r}$ makes an angle θ with the positive x-axis.
Component of $\vec{r}$ along x-axis

$$r_x = r \cos \theta$$

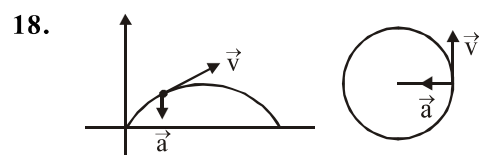
$$(r_x)_{\max} = r (\cos \theta)_{\max}$$

$$\therefore (\cos \theta)_{\max} = 1 \text{ at } \theta = 0^\circ$$
 hence $\vec{r}$ is along positive x-axis.
- In two dimensional motion, if instantaneous speed is a positive constant then equal path lengths are traversed in equal intervals of time.
- Three vectors of different magnitudes in the same plane can give the zero resultant. If the components of vector are not perpendicular to each other then the component of a vector may be greater than the magnitude of the vector. Combination of two vectors of different magnitudes can never give the zero resultant.
- Both marbles has zero initial velocity in vertical direction, same acceleration and same displacement in vertical direction. Hence both will reach the ground at the same instant.
- If λ is -ve then $|\vec{P}| = -\lambda |\vec{Q}|$, $\hat{P} = -\hat{Q}$
 Dimension of $\vec{P} = \text{Dim. of } \lambda \vec{Q}$
1. Option is correct
 2. Option is wrong value of average velocity in a given interval of time is a single value but instantaneous values of velocities between the same interval can be different.
 3. Option is wrong, in retardation direction of motion and acceleration are different.
 4. Option is wrong, if velocity and acceleration remain in same direction the motion of particle is a straight line.
1. Option is wrong if speed of particle changes then the direction of resultant acceleration is not towards the centre.
 2. Option is wrong.
 3. Option is correct
 4. Option is wrong, velocity is tangential.
- Perpendicular acceleration changes direction of motion. It cannot change speed.

14. 1. Option is wrong, acceleration is variable
2. Option is correct.
3. Option is wrong. Net workdone is zero is speed is constant
4. Option is wrong, tangential force deliver power because it can be parallel to velocity.
15. If distance > |displacement| then Average speed > |Average velocity|
16. All these are pre-conditions for projectile motion.

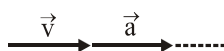


At any particular instant, direction of motion i.e. instantaneous velocity is tangential to curve (path).



In projectile and circular motion $\vec{a}$ and $\vec{v}$ are not in same direction.

In one dimensional motion $\vec{v}$ & $\vec{a}$ may have same direction.



19. 1. Option is correct
2. Option is correct
3. Option is correct
4. Option must be incorrect, it is possible only if $\vec{R}$ & $\vec{S}$ are collinear.
20. The motion of the train only affects the magnitude of horizontal component of velocity and the acceleration of the body along the horizontal, if any. It does not affect the nature of the motion along vertical. Therefore in all three cases, the time of free fall of the body will be equal.
21. Range = $\frac{u^2 \sin 2\theta}{g}$. It is clear that range is proportional to the direction (angle) and the initial speed.
22. Change in momentum of the ball
 $= m u \sin \theta - (-m u \sin \theta) = 2m u \sin \theta$
 $= mg \frac{2u \sin \theta}{g} = \text{weight} \times \text{total time of flight}$

23. At the topmost point.
24. Because vertical downward displacement of both (target and bullet) will be equal.

25. (i) Time of flight = $\frac{2u \sin \theta}{g} = \frac{2(u_{\text{vertical}})}{g}$

26. $V \cos \theta = U \cos (\theta)$

$$V = \frac{U \cos 2\theta}{\cos \theta}$$

$$V = \frac{U[2 \cos^2 \theta - 1]}{\cos \theta} = U[2 \cos \theta - \sec \theta]$$

In projectile motion, horizontal component of velocity remains constant.

27. downward open  negative curvature
upward open  positive curvature

28. Because horizontal velocity is same for coin and the observer. So relative horizontal displacement is zero.
29. Due to air resistance, its horizontal velocity will decrease so it will fall behind the aeroplane.
30. At highest point of trajectory of projectile tangential component of acceleration is zero as acceleration is vertically downwards (g) and velocity is horizontal or the angle between acceleration and velocity is 90° .
31. Velocity is tangential and acceleration is radial.
32. Speed of particle is increasing, means tangential component of acceleration is present and is along the velocity of particle. So, dot product of acceleration and velocity of particle is positive.